

MAT1575, Classwork4, Fall2026

1.3 The Fundamental Theorem of Calculus (p.40-56)

1. **Definition:** Average value of the function.

Let $f(x)$ be continuous on the interval $[a, b]$. Then, the average value of the function $f(x)$, denoted f_{ave} , on $[a, b]$ is given by

$$f_{ave} = \frac{1}{b-a} \int_a^b f(x) dx$$

2. **Example:** Find the average value of $f(x) = 6 - 2x$ on the interval $[0, 3]$.

$$f_{ave} = \frac{1}{3-0} \int_0^3 (6-2x) dx = \frac{1}{3} \int_0^3 (6-2x) dx = ? \quad \frac{1}{3} [6x - x^2]_0^3$$

3. **Theorem: The Mean Value Theorem for Integrals.**

If $f(x)$ is continuous on interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx \quad \text{or} \quad \int_a^b f(x) dx = f(c)(b-a)$$

4. **Example.** Find the average value of function $f(x) = \frac{x}{2}$ over the interval $[0, 6]$ and find a $c \in [0, 6]$ equals the average value of the function over $[0, 6]$.

$$f_{ave} = \frac{1}{6-0} \int_0^6 \frac{x}{2} dx = \frac{1}{6} \left. \frac{x^2}{4} \right|_0^6 = \frac{36}{24} - 0 = \frac{3}{2} \quad f(c) = \frac{3}{2} \Rightarrow c = 3.$$

5. **Theorem. Fundamental Theorem of Calculus, Part 1.**

If $f(x)$ is continuous on interval $[a, b]$, and the function $F(x)$ is defined by

$$F(x) = \int_a^x f(t) dt, \text{ then } F'(x) = f(x) \text{ over } [a, b].$$

Proof: we have a $c \in [x, x+h]$ such that

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right] = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt = \lim_{h \rightarrow 0} f(c) = f(x)$$

since $c \rightarrow x$ as $h \rightarrow 0$

6. **Example.** Use the Fundamental Theorem of Calculus, Part 1 to find the derivative of

(a) $G(x) = \int_1^x \frac{1}{t^3+1} dt$
 $G'(x) = \frac{1}{x^3+1}$

(b) $F(r) = \int_0^r \sqrt{x^2+4} dx$
 $F'(r) = \sqrt{r^2+4}$

(c) $H(x) = \int_1^{\sqrt{x}} \sin(t) dt$
 $u(x) = \sqrt{x} = x^{\frac{1}{2}} \quad u'(x) = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$
 $H(x) = \int_1^{u(x)} \sin(t) dt$
 $H'(x) = \sin(u(x)) \cdot u'(x)$
 $= \sin(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$
 $= \frac{\sin(\sqrt{x})}{2\sqrt{x}}$

(d) $K(x) = \int_x^{2x} t^3 dt$
 $K(x) = \int_0^{2x} t^3 dt - \int_0^x t^3 dt$
 $K'(x) = (2x)^3 \cdot 2 - x^3 = 15x^3$
 $\frac{d}{dx}(2x)$

Observation: To make **Fundamental Theorem of Calculus, Part 1** work, we need to have

- The lower limit of integration has to be a constant
- The upper limit of integration has to be the variable matching with the variable of F , but not the variable of integration
- If the upper limit of integration is another function, then there will be extra term(s) from chain rule for F'

7. Theorem: Fundamental Theorem of Calculus, Part 2: The Evaluation Theorem

If $f(x)$ is continuous on interval $[a, b]$, and the function $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a). \quad \text{or } [F(x)]_a^b = F(b) - F(a)$$

Proof: Let $P = \{x_i\}$ for $i = 1, 2, \dots, n$ be a partition on $[a, b]$. By Mean Value Thm, we can find a $c_i \in [x_{i-1}, x_i]$

So, $\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) \Rightarrow F(x_i) - F(x_{i-1}) = F'(c_i)(x_i - x_{i-1}) = f(c_i)(x_i - x_{i-1})$

Then $F(b) - F(a) = \sum_{i=1}^n f(c_i) \Delta x \Rightarrow \lim_{n \rightarrow \infty} (F(b) - F(a)) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x = \int_a^b f(x) dx$

Take limit on both sides and get $F(b) - F(a) = \int_a^b f(x) dx$

observation: if we can find the antiderivative for the integrand, then we can evaluate the definite integral by evaluating the antiderivative at the endpoints of the interval and subtracting.

The drawback of this method: We must be able to find an antiderivative, and this is not always easy.

8. Evaluate $\int_{-2}^2 (t^2 - 4) dt = \left[\frac{t^3}{3} - 4t \right]_{-2}^2$

$t=2 \Rightarrow \left[\frac{(2)^3}{3} - 4 \cdot 2 \right] - \left[\frac{(-2)^3}{3} - 4 \cdot (-2) \right]$

$= \left[\frac{8}{3} - 8 \right] - \left[-\frac{8}{3} + 8 \right] = \frac{8}{3} - 8 + \frac{8}{3} - 8 = \frac{16}{3} - 16 = -\frac{32}{3}$

9. Evaluate $\int_1^9 \frac{y-1}{\sqrt{y}} dy = \int_1^9 \left(\frac{y}{\sqrt{y}} - \frac{1}{\sqrt{y}} \right) dy = \int_1^9 \left(y^{\frac{1}{2}} - y^{-\frac{1}{2}} \right) dy = \left[\frac{2}{3} y^{\frac{3}{2}} - 2y^{\frac{1}{2}} \right]_1^9$

$y^{\frac{3}{2}} = (\sqrt{y})^3$
 $y^{\frac{1}{2}} = \sqrt{y}$

$= \left[\frac{2}{3} (\sqrt{9})^3 - 2 \cdot \sqrt{9} \right] - \left[\frac{2}{3} (\sqrt{1})^3 - 2 \cdot \sqrt{1} \right]$

$= \left[\frac{2}{3} \cdot 27 - 2 \cdot 3 \right] - \left[\frac{2}{3} - 2 \right]$

$= 18 - 6 - \frac{2}{3} + 2 = 14 - \frac{2}{3} = \frac{40}{3}$

10. James skates along a long, straight track at a velocity of $f(t) = 10 + \cos\left(\frac{\pi}{2}t\right)$ ft/sec. Find how far James has gone after 5 seconds.

$$\int_0^5 10 + \cos\left(\frac{\pi}{2}t\right) dt = \left[10t + \frac{2}{\pi} \sin\left(\frac{\pi}{2}t\right) \right]_0^5$$

$$= \left[10 \cdot 5 + \frac{2}{\pi} \sin\left(\frac{\pi}{2} \cdot 5\right) \right] - \left[10 \cdot 0 + \frac{2}{\pi} \sin\left(\frac{\pi}{2} \cdot 0\right) \right]$$

$$= \left[50 + \frac{2}{\pi} \sin\left(\frac{5\pi}{2}\right) \right] - 0$$

$$\sin\left(\frac{5\pi}{2}\right) = \sin\left(\frac{5\pi}{2} - 2\pi\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

